The sign of the mass self-energy in the Born analysis of "Anomalous quantum Hall effect induced by disorder in topological insulators"

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Abstract

The self-consistent Born approximation (SCBA) used in Ref. [1], and in Chapter 2 of the thesis [2], renormalises the masses of the two spin blocks with the wrong sign. The error is in the definition of the renormalised (barred) parameters, $\mathrm{Re}\,\Sigma \equiv -\mathrm{diag}[(\bar{m}-m)\tau_z + (\bar{\mu}-\mu)\tau_0, \dots]$, which holds for the chemical potential but not for the mass. The program that computed the Born phase diagrams (Fig. 4 of Ref. [1]) implements these equations as written. With the correct sign, the disorder fluctuations push both masses towards the topological side, as in the topological Anderson insulator that the paper follows. Three checks confirm it, none of which uses the derivation: a full matrix SCBA, the exact disorder-averaged Green's function, and spin-resolved transport. The numerical results of the paper (Figs. 1–3) are unaffected, and so is its main conclusion: a $G = 1$ Chern-insulator phase, carried by the spin-down edge state, for both couplings. What changes is the Born interpretation. In the same-g ("ferromagnetic") case, the spin-down block, which carries the $G = 1$ plateau, never changes sign, whereas the published lines invert it. The spin-up channel is lost because the Fermi level enters the spin-up band, shifted by the mean exchange field. At stronger disorder the spin-up block does become trivial, as the paper says, but through a resonance that no Born approximation describes. In the opposite-g ("antiferromagnetic") case, the spin-up mass changes sign at $J_I \simeq 0.40$, not 0.30. The apparent agreement of the published inversion line with the antiferromagnetic conductance map is a finite-width effect of the 64-site ribbons.

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1 Summary

- **The error.** The mass equations of the SCBA carry the wrong sign (Sec. 3). The chemical-potential equations are right.
- **Where it comes from.** The derivation, as printed in Ref. [1] and in the thesis [2]: the code `self_born.f90` implements the printed equations faithfully. A replica of the program reproduces the published Born data (runs 00–07) to 10^{-14}.
- **Independent evidence** (Sec. 4):
  1. a matrix SCBA computed from the Hamiltonian, without the reduction;
  2. the exact renormalised masses, from the disorder-averaged Green's function (Fig. 1);
  3. the spin-resolved conductance of the same ribbons as the paper.

  All three follow the corrected sign. At weak disorder the corrected SCBA agrees with the exact masses to 10^{-3}, while the published one moves them in the opposite direction.
- **Impact on Fig. 4.** With the published maps (Fig. 2, Sec. 5):
  - same-g: the corrected lines explain 99.9 % of the $G = 1$ plateau, the published ones 20 %;
  - opposite-g: the published spin-up inversion line sits at the left edge of the $G = 1$ region, the corrected one inside it.
- **Why the published lines looked right** (Sec. 6). The ribbons are 64 sites wide. Close to the inversion, the spin-up edge states decay over $1.5/|\bar{m}_\uparrow|$ sites, so they overlap across the ribbon and the spin-up channel is lost before the mass changes sign. Wider ribbons move the transition towards the exact inversion. At $W = 256$ the spin-up block still carries $T_\uparrow = 0.99$ where the published masses make it trivial. The wrong sign moves the Born inversion by about the same amount in the same direction: a coincidence.
- **Origin and stability of $G = 1$** (Sec. 7). $G = 1$ is a Chern insulator: the spin-down block keeps its topological gap, while the spin-up states at the Fermi level no longer conduct. Each edge then carries a single chiral mode, with nothing to backscatter into. The same-g plateau does not depend on the ribbon width, and it survives tilted moments ($M_z = 0.8$), whereas $G = 2$ does not.

2 The model and the published equations

The two spin blocks of the BHZ model are, on the square lattice (lattice constant 1),

$$h_\sigma(\boldsymbol{k}) = -D_{\boldsymbol{k}}\,\tau_0 + (m - B_{\boldsymbol{k}})\,\tau_z + A\,(\text{terms odd in } \boldsymbol{k}), \qquad B_{\boldsymbol{k}} = B(4 - 2\cos k_x - 2\cos k_y), \quad (1)$$

and similarly for $D_{\boldsymbol{k}}$. The parameters are $A = 1$, $B = -0.376406$, $D = -0.280933$ and $m = -0.137$, so the clean system is topological, $m/B > 0$. The Pauli matrices τ act on the electron (E) and hole (H) orbitals.

The magnetic impurities occupy a fraction n_I of the sites. They carry classical moments $\boldsymbol{S}_i$ with $\langle S_z \rangle = M_z$ and couple through

$$V_\alpha = J_I \sum_{i \in I} \boldsymbol{S}_i \cdot \boldsymbol{\sigma}\, \tau_\alpha, \qquad \tau_\alpha = \tau_0 \text{ (same-}g\text{, “F”)}, \quad \tau_\alpha = \tau_z \text{ (opposite-}g\text{, “AF”)}. \tag{2}$$

This note treats the fully polarised case $M_z = 1$ of the paper's Figs. 1, 2 and 4. Then $V_\alpha = J_I \sigma_z \tau_\alpha$ on the impurity sites, s_z is conserved, and the two spin blocks are independent disordered systems. The mean field $\langle V_\alpha \rangle = n_I J_I M_z\, \sigma_z \tau_\alpha$ shifts, block by block:

- the chemical potential, $\mu_\sigma = \mu - \sigma n_I J_I M_z$, for same-$g$;
- the mass, $m_\sigma = m + \sigma n_I J_I M_z$, for opposite-$g$.

Ref. [1] (Secs. IV A “Born approximation” and IV B “Phase diagram”) and the thesis [2] (Sec. “Discussion : l'approximation de Born”) write the mean-field Green's function and the SCBA self-energy as

$$g_\alpha(E, \boldsymbol{k}) = \left[E + \mu - H_0(\boldsymbol{k}) - \langle V_\alpha \rangle\right]^{-1}, \qquad \Sigma(E) = \frac{n_I J_I^2}{N^2} \sum_{\boldsymbol{k}} \left[v_z\, g_z + v_\perp g_\perp\right], \tag{3}$$

where g is replaced by the full averaged Green's function in the self-consistent version. They then *define* the renormalised parameters by

$$\mathrm{Re}\,\Sigma \equiv -\mathrm{diag}\big[(\bar m_\uparrow - m_\uparrow)\tau_z + (\bar\mu_\uparrow - \mu_\uparrow)\tau_0,\ (\bar m_\downarrow - m_\downarrow)\tau_z - (\bar\mu_\downarrow - \mu_\downarrow)\tau_0\big] \qquad \text{(as printed)}, \tag{4}$$

and obtain

$$\bar\mu_\sigma = \mu_\sigma - \frac{n_I J_I^2}{N^2} \sum_{\boldsymbol{k}} \Big[\frac{v_z}{\Delta_\sigma}(E_F + \bar\mu_\sigma + D_{\boldsymbol{k}}) + \frac{v_\perp}{\Delta_{\bar\sigma}}(E_F + \bar\mu_{\bar\sigma} + D_{\boldsymbol{k}})\Big], \tag{5}$$

$$\bar m_\sigma = m_\sigma - \frac{n_I J_I^2}{N^2} \sum_{\boldsymbol{k}} \Big[\frac{v_z}{\Delta_\sigma}(\bar m_\sigma - B_{\boldsymbol{k}}) + \frac{v_\perp}{\Delta_{\bar\sigma}}(\bar m_{\bar\sigma} - B_{\boldsymbol{k}})\Big], \tag{6}$$

with $\Delta_\sigma = (E_F + \bar\mu_\sigma + D_{\boldsymbol{k}})^2 - (\bar m_\sigma - B_{\boldsymbol{k}})^2 - \sin^2 k_x - \sin^2 k_y$. The gap of block σ is centred on $-\bar\mu_\sigma$, with half-width $|\bar m_\sigma|$, and the block is topological when $\bar m_\sigma / B > 0$. Fig. 4 of Ref. [1] draws the band edges $E_F = -\bar\mu_\sigma \pm \bar m_\sigma$ and the lines $\bar m_\sigma = 0$.[1]

The program `self_born.f90`, which produced Fig. 4, iterates exactly Eqs. (5) and (6). In it, `norm` is $n_I J_I^2/N^2$, `kc` is $B_{\boldsymbol{k}}/B$, and `vzu = vz/detu` with `detu` $= \Delta_\uparrow$:

```
154  vz = (1-conc**2)*Mz**2 + (1-Mz)**2/3.0
192  deltaMu = -norm * sum( vzu*(Mu - B*kc) + vperpd*(Md - B*kc) )
200  Mu_temp = Mu0 + deltaMu
```

A replica written for this note reproduces the published Born data of runs 00–07 to 10^{-14}. The code is therefore a faithful implementation of the printed equations.

[1] Eq. (4) as printed also has opposite signs for the τ_0 terms of the two blocks. Eqs. (5) use the same sign for both, which is the right one. The thesis writes $\mu_\uparrow$ instead of $m_\uparrow$ on the right-hand side of the spin-up mass equation; the program uses $m_\uparrow$.

3 The error

3.1 Derivation

The averaged Green's function is $\langle G\rangle = [E+\mu - H_0 - \langle V\rangle - \Sigma]^{-1}$. The renormalised parameters are defined by writing it as the Green's function of a clean block with barred parameters:

$$E + \mu_\sigma - h_\sigma(\boldsymbol{k}; m_\sigma) - \Sigma_\sigma = E + \bar{\mu}_\sigma - h_\sigma(\boldsymbol{k}; \bar{m}_\sigma). \tag{7}$$

The mass enters h_σ as $+m\,\tau_z$ (Eq. 1), the chemical potential as $-\mu\,\tau_0$. Identifying the τ_0 and τ_z components gives

$$\boxed{\operatorname{Re}\Sigma_\sigma = (\bar{m}_\sigma - m_\sigma)\,\tau_z - (\bar{\mu}_\sigma - \mu_\sigma)\,\tau_0} \tag{8}$$

The minus sign of Eq. (4) is right for μ and wrong for m. The τ_z component of g_σ is $(\bar{m}_\sigma - B_{\boldsymbol{k}})/\Delta_\sigma$ (Ref. [1], Eq. for g_z). The mass equation therefore reads

$$\boxed{\bar{m}_\sigma = m_\sigma + \frac{n_I J_I^2}{N^2}\sum_{\boldsymbol{k}}\Big[\frac{v_z}{\Delta_\sigma}(\bar{m}_\sigma - B_{\boldsymbol{k}}) + \frac{v_\perp}{\Delta_{\bar{\sigma}}}(\bar{m}_{\bar{\sigma}} - B_{\boldsymbol{k}})\Big]} \tag{9}$$

Eq. (5) for $\bar{\mu}_\sigma$ is unchanged.

3.2 One-line check

Treat a uniform, non-random term $\delta\,\tau_z$ as a self-energy: $\Sigma = \delta\,\tau_z$ exactly, and it shifts the mass to $m+\delta$. Eq. (4) gives $\bar{m} = m - \delta$.

3.3 It is not a sign convention

Some authors write $\langle G\rangle = [g^{-1} + \Sigma]^{-1}$ with $\Sigma = -\langle VgV\rangle$. A change of convention flips the τ_0 and τ_z components of Σ together. Eqs. (5) and (6) treat them differently: both carry a minus sign in front of the sum, while μ enters the Hamiltonian with a minus sign and m with a plus sign. No convention makes both equations right; the checks of Sec. 4 show which one is wrong.

3.4 Consistency with the perturbative result and with the topological Anderson insulator

At lowest order, Ref. [1] obtains

$$\operatorname{Re}\Sigma = \frac{n_I J_I^2}{8\pi}\,\frac{B\tau_z - D\tau_0}{B^2 - D^2}\Big[v_z \log\Big|\frac{(B^2-D^2)k_*^4}{\Delta(E_F,0)}\Big| + \dots\Big], \tag{10}$$

with a large positive logarithm when E_F is in the gap. With Eq. (8), the mass correction $\bar{m} - m = \operatorname{Re}\Sigma_{\tau_z}$ has the sign of B. It moves $\bar{m}$ away from zero on the topological side, $\bar{m}/B > 0$. This is the renormalisation that explains the topological Anderson insulator [3, 4, 5], the mechanism the paper says it follows. With Eq. (4), the published equations push the masses towards the trivial side instead.

Physically, at second order the two couplings act alike on each block: τ_0 disorder (same-g) and τ_z disorder (opposite-g) give the same $\langle VgV\rangle$ once the terms odd in $\boldsymbol{k}$ are summed out, since $\tau_z(a + b\tau_z)\tau_z = a + b\tau_z$. Both are the scalar disorder of the topological Anderson insulator.

3.5 A second, smaller correction

The longitudinal weight is printed as $v_z = \langle S_z^2\rangle - n_I^2 M_z^2 = (1-n_I^2)M_z^2 + (1-M_z)^2/3$. The prefactor n_I of Eq. (3) is already factored out of the second cumulant of the Bernoulli occupation, $n_I\langle S_z^2\rangle - n_I^2 M_z^2$. The weight should therefore be $v_z = (1-n_I)M_z^2 + (1-M_z)^2/3$. The printed value overestimates the longitudinal fluctuations by a factor $1+n_I$. This factor is positive, so it changes the size of the fluctuation term and never its direction: the direction is set by the sign alone. The corrected curves below include both corrections.

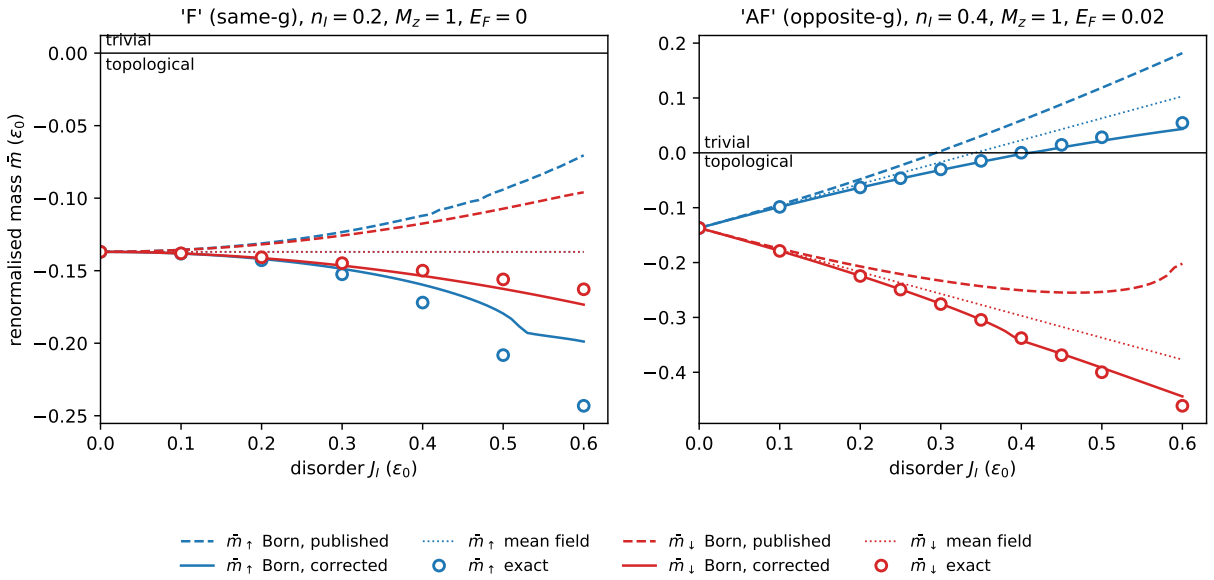


Figure 1: Renormalised masses: exact (circles, disorder-averaged Green's function, no Born approximation), published SCBA (dashed), corrected SCBA (solid), mean field (dotted). A block is topological for $\bar{m} < 0$. Left: same-g, $n_I = 0.2$, $E_F = 0$. Right: opposite-g, $n_I = 0.4$, $E_F = 0.02$.

4 Checks that do not use the derivation

4.1 Matrix SCBA

The SCBA can be solved without the reduction to four scalars. The self-energy is iterated as a 4×4 matrix, $\Sigma = \sum_t w_t V_t \langle G\rangle_{\mathrm{loc}} V_t$, where V_t are the exchange vertices with their second-cumulant weights and $\langle G\rangle_{\mathrm{loc}} = N^{-2}\sum_{\boldsymbol{k}}[E + \mathrm{i}\eta - H_0(\boldsymbol{k}) - \langle V\rangle - \Sigma]^{-1}$, on the same k grid. The reduced equations with the sign of Eq. (9) reproduce this matrix solution to 10^{-7}, for both couplings. With the sign of Eq. (6) they do not.

4.2 Exact renormalised masses

The disorder-averaged Green's function at $\boldsymbol{k} = 0$ defines the exact self-energy, to all orders in J_I:

$$\Sigma(E) = E - H_0(\mathbf{0}) - \langle G(\mathbf{0}, E)\rangle^{-1}. \tag{11}$$

In each block, $H_0(\mathbf{0}) = m\tau_z$ and $\langle G(\mathbf{0})\rangle$ is diagonal in the E/H basis, so

$$\bar{m} = m + \tfrac{1}{2}\,\mathrm{Re}(\Sigma_{EE} - \Sigma_{HH}), \qquad \bar{\mu} = -\tfrac{1}{2}\,\mathrm{Re}(\Sigma_{EE} + \Sigma_{HH}), \qquad \Gamma = -\tfrac{1}{2}\,\mathrm{Im}(\Sigma_{EE} + \Sigma_{HH}). \tag{12}$$

These include the mean field, like the SCBA's $\bar{m}_\sigma$ and $\bar{\mu}_\sigma$. The broadening Γ stays near zero when no state lies at E.

We computed $\langle G(\mathbf{0})\rangle_{aa} = \langle\phi_a|(E + \mathrm{i}\eta - H)^{-1}|\phi_a\rangle$:

- ϕ_a is the normalised $\boldsymbol{k} = 0$ plane wave on orbital a;
- H is the Hamiltonian of a 160×160 periodic lattice with $\lfloor n_I \cdot 160^2 \rfloor$ impurities, and $\eta = 0.006$;
- the resolvent is expanded in Chebyshev polynomials of the rescaled Hamiltonian (Appendix A).

Two independent samples agree to 10^{-3}. Fig. 1 and Table 1 compare the result with both versions of the SCBA.

Table 1: Renormalised masses, exact and in the two versions of the SCBA.

case	J_I	exact	published	corrected
same-g, $\bar{m}_\uparrow$, $E_F = 0$	0.2	-0.1428	-0.1312	-0.1419
	0.4	-0.1720	-0.1121	-0.1597
	0.6	-0.2432	-0.0705	-0.1989
same-g, $\bar{m}_\downarrow$, $E_F = 0$	0.2	-0.1408	-0.1319	-0.1413
	0.4	-0.1500	-0.1175	-0.1536
	0.6	-0.1628	-0.0960	-0.1734
opposite-g, $\bar{m}_\uparrow$, $E_F = 0.02$	0.30	-0.0302	$+0.0029$	-0.0313
	0.35	-0.0147	$+0.0307$	-0.0164
	0.40	$+0.0002$	$+0.0591$	-0.0021
	0.45	$+0.0145$	$+0.0884$	$+0.0098$
opposite-g, $\bar{m}_\downarrow$, $E_F = 0.02$	0.40	-0.3377	-0.2506	-0.3422
same-g, $\bar{m}_\downarrow$, $E_F = -0.05$	0.8	-0.1790	-0.0605	-0.2084
	1.0	-0.1964	-0.0176	-0.2569
	1.2	-0.2150	$+0.0331$	-0.3129
	1.5	-0.2485	$+0.1235$	-0.3511

- **Direction.** Measured from the mean field, the exact masses move towards the topological side, like the corrected SCBA and unlike the published one.
- **Weak disorder.** At same-g, $J_I = 0.2$, the corrected SCBA is within 10^{-3} of the exact masses. The published one is off by 0.01, in the opposite direction.
- **The inversion.** At opposite-g, the exact spin-up mass changes sign at $J_I = 0.40$, where the corrected SCBA puts it. The published SCBA puts it at 0.30.
- **Strong disorder.** The corrected SCBA remains close up to $J_I = 0.6$. At same-g the exact masses are even more topological (higher orders).
- **The block that carries $G = 1$.** At same-g and $E_F = -0.05$, deep in the $G = 1$ plateau, the exact spin-down mass stays topological and its gap widens up to $J_I = 1.5$. The broadening there stays below 0.005: no state lies at E_F. The published SCBA inverts this block at $J_I \approx 1.1$, which does not happen.
- **Spin up at strong disorder.** Table 2 follows the line $E_F = -0.06$, through the onset of the $G = 1$ plateau.
  - For $J_I \gtrsim 0.4$, E_F lies in the spin-up band, and the spin-up states are strongly scattered.
  - Near $J_I \approx 0.87$, the $\boldsymbol{k} = 0$ self-energy of that block goes through a resonance: the broadening peaks at 1.2, and the mass diverges and changes sign.
  - Beyond it, the block has a trivial gap around E_F: $\bar{m}_\uparrow > 0$, with a broadening of 0.01 at $J_I = 1.2$ and 0.006 at 1.5 ($E_F = -0.05$).

  This is a strong-scattering effect, outside any Born approximation. The published SCBA reaches a trivial spin-up block through a smooth inversion, at $J_I = 0.72$ on this line; the corrected one never does. Both predict $T_\uparrow = 0$ in this region, as observed. The spin-down block stays topological throughout, with no state at E_F.

Table 2: Exact $\boldsymbol{k} = 0$ parameters along $E_F = -0.06$ (same-g, $n_I = 0.2$), across the onset of the $G = 1$ plateau, with the published spin-up mass for comparison.

J_I	$\bar{m}_\uparrow$	$\Gamma_\uparrow$	published $\bar{m}_\uparrow$	$\bar{m}_\downarrow$	$\Gamma_\downarrow$
0.60	-0.270	0.10	-0.055	-0.164	0.0004
0.70	-0.273	0.21	-0.010	-0.171	0.0005
0.80	-0.569	0.37	$+0.053$	-0.179	0.0006
0.85	-1.212	1.24	$+0.088$	-0.183	0.0006
0.90	$+1.176$	0.38	$+0.113$	-0.188	0.0007
0.95	$+0.552$	0.08	$+0.124$	-0.192	0.0008
1.00	$+0.369$	0.04	$+0.132$	-0.197	0.0008

Table 3: Exact spin-resolved transmissions and the renormalised mass of the relevant block in the two versions of the SCBA. Means over 8 configurations ($W = 64$, 128; ribbons of 400 sites) or 3 configurations ($W = 256$; disordered length 256).

case	J_I	E_F	$T_\uparrow$	$T_\downarrow$	published	corrected
same-g, $W = 64$	1.2	-0.05	0.000	1.000	$\bar{m}_\downarrow = +0.033$ (trivial)	$\bar{m}_\downarrow = -0.313$
same-g, $W = 64$	1.3	-0.12	0.000	1.000	$\bar{m}_\downarrow = +0.080$ (trivial)	$\bar{m}_\downarrow = -0.314$
opposite-g, $W = 64$	0.30	0.02	0.840	1.000	$\bar{m}_\uparrow = +0.003$	$\bar{m}_\uparrow = -0.031$
opposite-g, $W = 64$	0.35	0.02	0.005	1.000	$\bar{m}_\uparrow = +0.031$ (trivial)	$\bar{m}_\uparrow = -0.016$
opposite-g, $W = 128$	0.30	0.02	0.997	1.000		
opposite-g, $W = 128$	0.35	0.02	0.562	1.000		
opposite-g, $W = 128$	0.40	0.02	0.012	1.000	$\bar{m}_\uparrow = +0.059$	$\bar{m}_\uparrow = -0.002$
opposite-g, $W = 256$	0.35	0.02	0.989	1.000	$\bar{m}_\uparrow = +0.031$ (trivial)	$\bar{m}_\uparrow = -0.016$
opposite-g, $W = 256$	0.40	0.02	0.339	1.000	$\bar{m}_\uparrow = +0.059$	$\bar{m}_\uparrow = -0.002$

4.3 Spin-resolved transport

With $M_z = 1$, the transmission splits exactly into $T = T_\uparrow + T_\downarrow$. We computed it on the ribbons of the paper's maps: 400×64 sites, $n_I \cdot 256 \cdot 64$ impurities in the central 256 columns, 8 configurations per point. Table 3 lists the points where the two versions disagree on the topology of a block.

- **Same-g.** The spin-down block carries a quantized channel, in 8 configurations out of 8, where the published masses make it trivial. A trivial block cannot carry a quantized chiral channel.
- **Opposite-g.** The spin-up channel vanishes earlier than the exact inversion, and later when the ribbon is wider. A trivial block would carry no channel at any width. This finite-width effect is the subject of Sec. 6.

5 Impact on the Born phase diagrams

Fig. 2 draws both versions of the Born lines over the published conductance maps. These are the data of Fig. 1 of Ref. [1], averaged over 101 configurations (opposite-g) and 50 configurations (same-g), in the colours of the thesis. Table 4 counts the points of the $G = 1$ plateau that each version explains. A block counts for one channel if it is topological and E_F lies in its gap ("localised" reading). Alternatively it counts if it is topological, whatever E_F ("topological" reading).

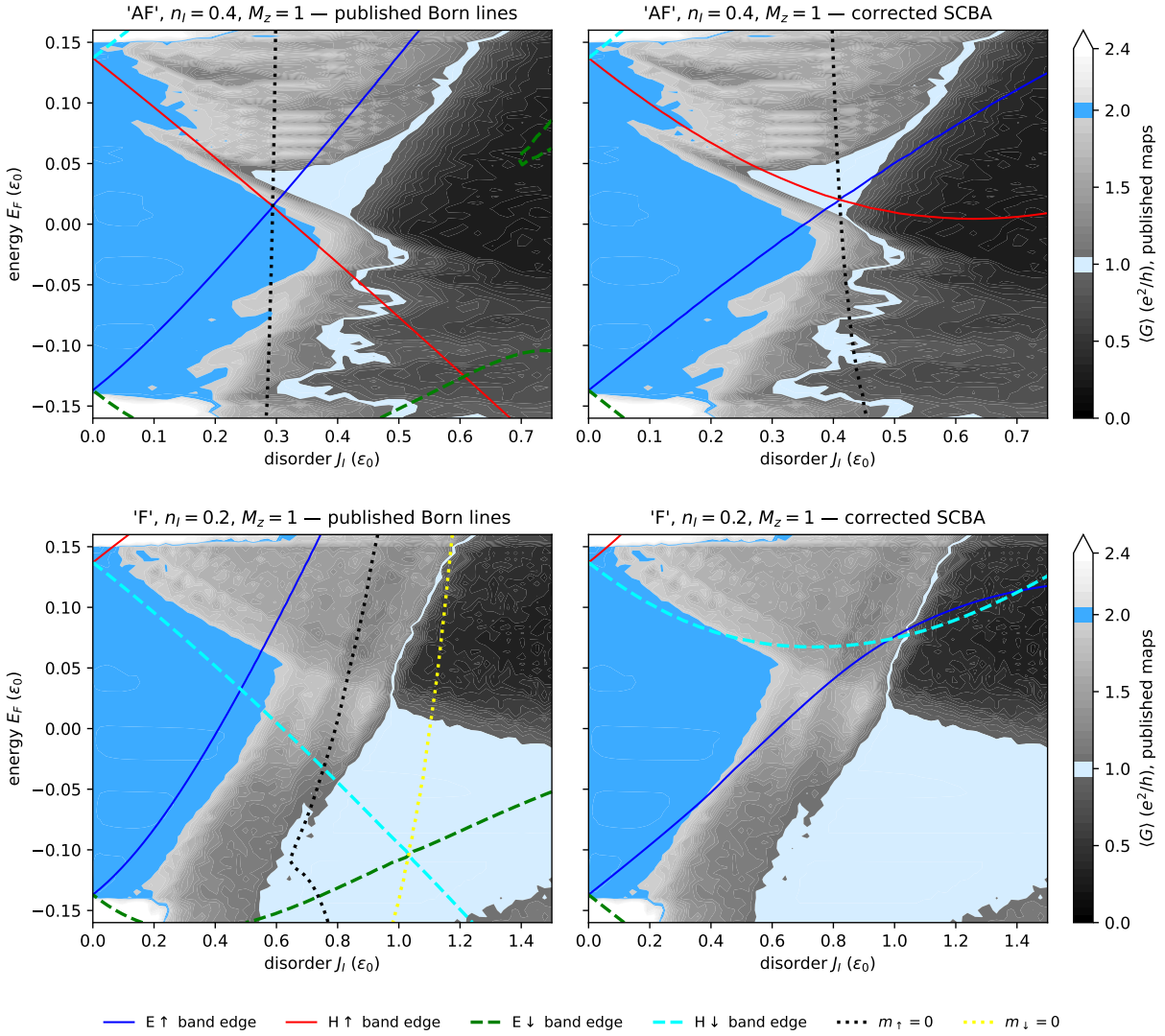


Figure 2: Born lines over the published conductance maps (Fig. 1 of Ref. [1]): published (left) and corrected SCBA (right). Light blue: $|\langle G \rangle - 1| < 0.05$; blue: $|\langle G \rangle - 2| < 0.05$. Solid lines: spin-up band edges; dashed: spin-down band edges; dotted: $\bar{m}_\sigma = 0$.

Table 4: Share of the $G = 1$ plateau ($|\langle G\rangle - 1| < 0.05$) of the published maps predicted at $G = 1$ by the Born lines.

		localised reading		topological reading	
map	$G = 1$ points	published	corrected	published	corrected
same-g, $n_I = 0.2$	1065	20 %	99.9 %	39 %	0 %
opposite-g, $n_I = 0.4$	122	94 %	88 %	96 %	59 %

Table 5: Same-g published map along lines of constant E_F: where $\langle G\rangle$ leaves 2 (< 1.95) and where the $G = 1$ plateau begins ($|\langle G\rangle - 1| < 0.05$), against the corrected spin-up band edge and the published $\bar{m}_\uparrow = 0$ line.

E_F	G leaves 2	plateau begins	corrected E↑ edge	published $\bar{m}_\uparrow = 0$
−0.109	0.24	0.62	0.14	0.65
−0.084	0.31	0.64	0.26	0.68
−0.058	0.38	0.76	0.38	0.72
−0.033	0.48	0.83	0.48	0.75
−0.008	0.52	0.93	0.59	0.78
+0.018	0.57	0.98	0.70	0.81

Same-g.

- **Published lines.** Both masses change sign: $\bar{m}_\uparrow = 0$ at $J_I \approx 0.65$–0.95, then $\bar{m}_\downarrow = 0$ at $J_I \approx 1.0$–1.2. Beyond the second line no block is topological, so G should vanish. The published map shows $G = 1$ there, up to $J_I = 1.5$: 566 of its 1065 $G = 1$ points lie where the published masses make both blocks trivial. The shaded $G = 1$ region of the paper's Fig. 4 covers only a small part of the plateau.
- **Corrected lines.** No mass changes sign. They locate where the spin-up channel starts to fall, at the spin-up band edge, and keep spin down topological, as the exact masses do. The strong-disorder end of the spin-up behaviour (Sec. 4.2) is beyond them.

Table 5 follows the published map along lines of constant E_F:

- G leaves 2 at the corrected spin-up band edge.
- The plateau begins 0.2–0.4 further, along a line with the same slope.
- The published $\bar{m}_\uparrow = 0$ line is almost vertical ($J_I = 0.65$–0.83). It lies within 0.04 of the onset of the plateau for $-0.11 \lesssim E_F \lesssim -0.06$, and 0.08–0.2 away elsewhere.
- On the line $E_F = -0.06$, the exact spin-up block does turn trivial, at $J_I \approx 0.9$, through the resonance of Sec. 4.2. The paper's picture of a trivial spin-up block therefore holds at strong disorder, but not through the mechanism, nor at the place, that the published line gives.

Opposite-g.

- **Published line.** $\bar{m}_\uparrow = 0$ at $J_I \approx 0.29$ lies at the left edge of the $G = 1$ region, as the paper describes.
- **Corrected line.** It lies at $J_I \approx 0.41$–0.45, inside that region.

Sec. 6 shows that the left edge of the $G = 1$ region is not the inversion but a finite-width effect.

6 Why the published inversion line matched the antiferromagnetic map

6.1 Overlap of the edge states

An edge state of a BHZ block with mass m decays into the bulk over $\xi = 1/\lambda_-$. Here $\lambda_\pm^2$ are the roots of $(E_0 - D\lambda^2)^2 = (m + B\lambda^2)^2 - A^2\lambda^2$ at the edge-state energy $E_0 = -mD/B$ [6]. For $|m| \ll A^2|B|/(B^2 - D^2)$,

$$\xi \simeq \frac{A|B|}{|m|\sqrt{B^2 - D^2}} \simeq \frac{1.50}{|m|} \text{ sites.} \tag{13}$$

This gives $\xi = 10.7$ sites for the clean mass, 50 sites for $|m| = 0.03$ and 100 sites for $|m| = 0.015$. In a ribbon of width W, the two counter-propagating spin-up edge states overlap once ξ becomes comparable with W. The disorder then backscatters between them across the ribbon, and the block's channel is lost while the block is still topological [6]. For the paper's $W = 64$, this happens at $|\bar{m}_\uparrow| \approx 0.015$–$0.03$ ($\xi \approx 50$–100 sites), i.e. $J_I \approx 0.30$–0.35 at $n_I = 0.4$ (Table 1). That is where the published map leaves $G = 2$, where the published $\bar{m}_\uparrow = 0$ line happens to lie.

6.2 Wider ribbons

Fig. 4 (left) follows both blocks along $E_F = 0.02$ for $W = 64$, 128 and 256. The disordered length is 256, with 8, 3 and 3 configurations.

- **The spin-up channel disappears later in wider ribbons.** $T_\uparrow = 1/2$ at $J_I \approx 0.32$ ($W = 64$), 0.355 ($W = 128$) and 0.39 ($W = 256$). The transition converges on the exact inversion, $J_I = 0.40$ (Table 1), and the channel is lost when ξ becomes comparable with W.
- **The published inversion is ruled out.** At $W = 256$ and $J_I = 0.35$, the spin-up block carries $T_\uparrow = 0.989$, the same within 0.002 in the three configurations. The published SCBA makes this block trivial there, with a gap of 0.06. A trivial block carries no channel at any width.
- **The spin-down channel also survives longer.** At $W = 256$, $T_\downarrow = 1.000$ up to $J_I = 0.475$, and it stays above 0.8 on average up to 0.6. At $W = 64$ it is already 0.89 at $J_I = 0.45$ and 0.60 at 0.5. Its loss in narrow ribbons comes from the impurity states that fill its gap (Sec. 7), which connect the two edges more easily when they are close.
- **The opposite-g $G = 1$ window moves and widens.** At $W = 256$ it starts at the exact inversion, $J_I \approx 0.40$, and extends beyond $J_I = 0.6$.

Fig. 4 (right) follows the same-g line $E_F = -0.06$.

- The spin-down block carries $T_\downarrow = 1.000$ at every J_I up to 1.5 and at every width, far beyond the published $\bar{m}_\downarrow = 0$ line.
- $T_\uparrow$ starts to fall at the corrected spin-up band edge ($J_I \approx 0.38$) at every width. It then crosses a diffusive region, where its value grows with W, and vanishes from $J_I \approx 0.8$ at every width.

Fig. 3 shows the full maps. The caroli maps at $W = 64$ reproduce the published ones. At $W = 128$:

- the same-g plateau is unchanged;
- the opposite-g $G = 2$ region extends to the corrected inversion, and the $G = 1$ region sits in the wedge beyond it, where the spin-up gap is trivial.

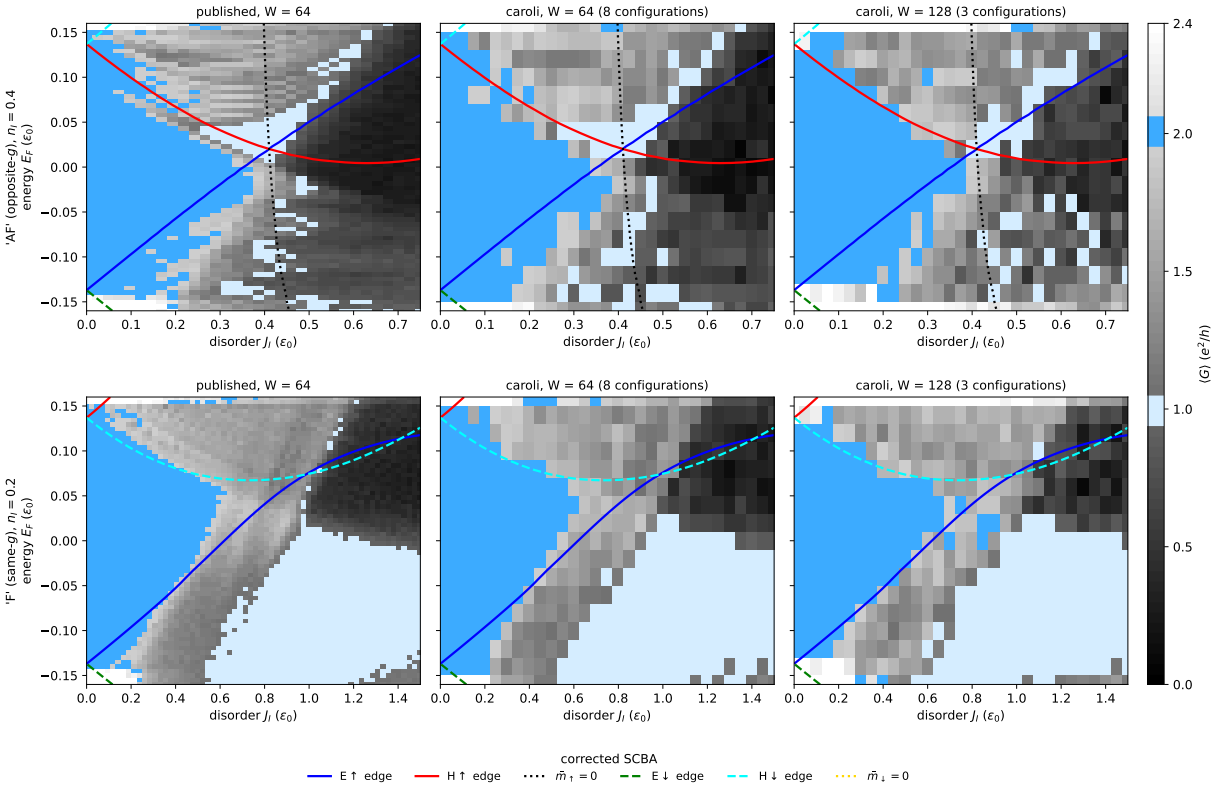


Figure 3: $\langle G \rangle$ maps in the colours of the thesis: published map ($W = 64$, Fortran, 101 and 50 configurations), and caroli on the same disordered length for $W = 64$ (8 configurations) and $W = 128$ (3 configurations). Lines: corrected SCBA.

6.3 The coincidence

The wrong sign moves the Born inversion from $J_I = 0.40$ to 0.30. The finite width moves the observed loss of the spin-up channel from 0.40 to about 0.32–0.35 at $W = 64$. The two shifts have the same direction and similar sizes, so the published line happened to fall on the edge of the published map. The same wrong sign predicts that both blocks become trivial at same-g, which the published map itself contradicts (Sec. 5).

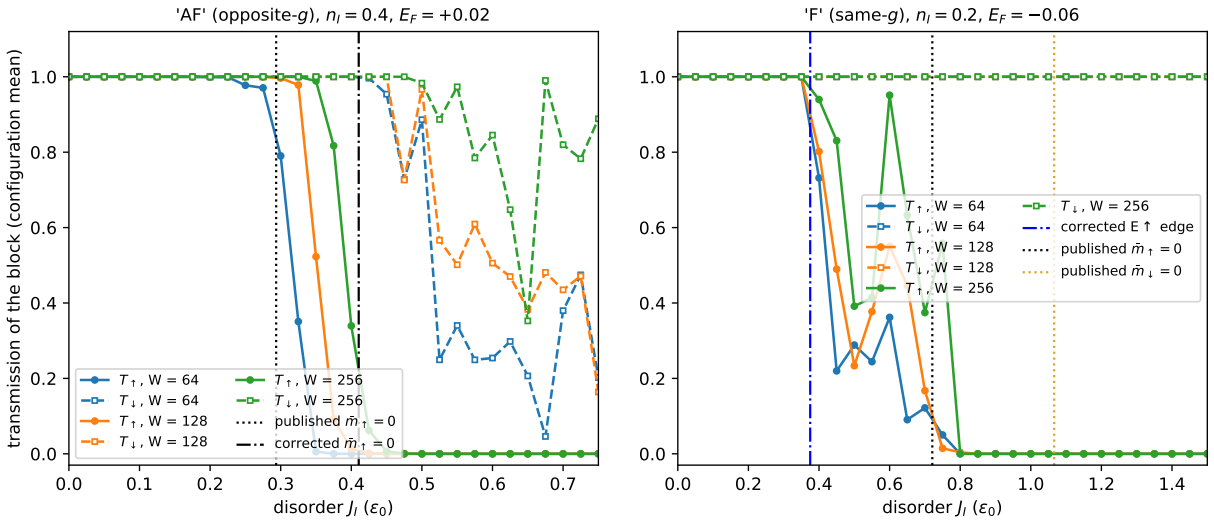


Figure 4: Transmissions of the two blocks along a line of constant E_F for $W = 64$, 128 and 256 (8, 3 and 3 configurations; disordered length 256). Left: opposite-g, $E_F = 0.02$, with the published and corrected spin-up inversions. Right: same-g, $E_F = -0.06$, with the corrected spin-up band edge and the published inversions of both blocks.

7 Origin and stability of the $G = 1$ phase

Fig. 5 resolves the conductance by spin. In each block, the corrected band edges of that block bound the region where it carries one channel.

Two independent Chern insulators. With $M_z = 1$, s_z is conserved and each block is a two-dimensional system without time-reversal symmetry (class A). Its two-terminal conductance is quantized to the Chern number of the occupied states whenever the states at E_F are localised. A block therefore contributes one channel as long as E_F lies above the delocalised states of its valence band and below those of its conduction band.

Same-g: how the phase appears.

- The mean exchange field acts as a spin-dependent chemical potential: the spin-up bands rise by $n_I J_I$ and the spin-down bands sink by $n_I J_I$.
- The fluctuations widen both gaps (Fig. 1).
- Near $E_F = 0$, the Fermi level enters the spin-up valence band as J_I grows: $T_\uparrow$ starts to fall at the corrected spin-up band edge.
- Once E_F lies below the delocalised spin-up states, the spin-up states at E_F are localised and carry no Chern number, so $T_\uparrow = 0$. The grey crossover of the map is the passage through those states.
- At stronger disorder ($J_I \gtrsim 0.9$ on $E_F = -0.06$), the spin-up block goes through a resonance and becomes a trivial insulator at E_F (Table 2). $T_\uparrow$ stays zero.
- The spin-down gap still contains E_F, so $T_\downarrow = 1$.
- The plateau ends only when E_F reaches the spin-down conduction band, at the top right of the map.

On the plateau of the $W = 64$ maps (8 configurations), $T_\downarrow$ is within 10^{-3} of 1 in 97 % of the single configurations, and $T_\uparrow < 10^{-3}$ in 92 %.

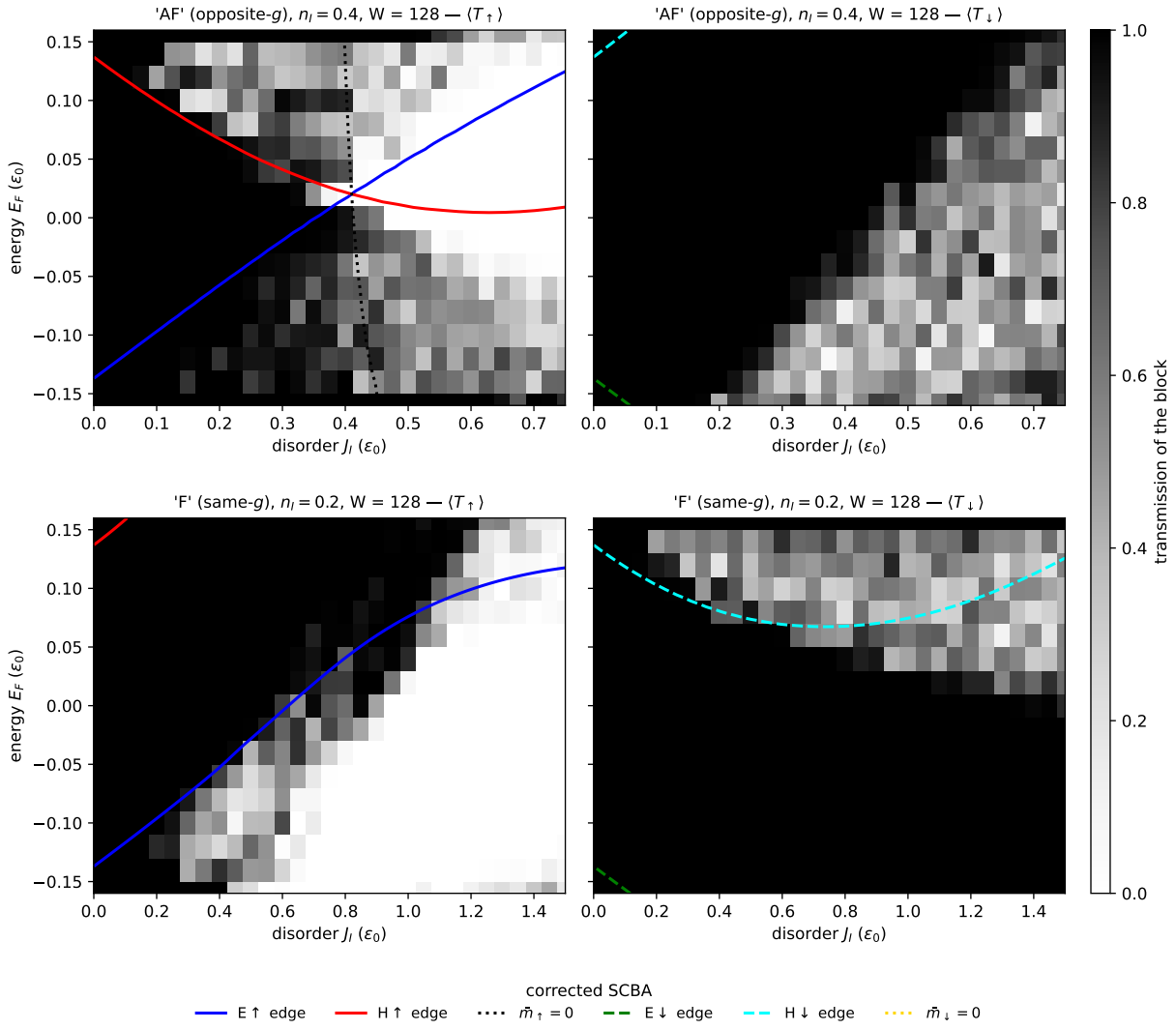


Figure 5: Spin-resolved transmissions $\langle T_\uparrow \rangle$ (left) and $\langle T_\downarrow \rangle$ (right), $W = 128$, 3 configurations, with the corrected band edges of each block. Black: the block carries one channel; white: none. Top: opposite-g, $n_I = 0.4$. Bottom: same-g, $n_I = 0.2$.

Why it is so stable.

- The $G = 1$ phase is a Chern insulator of total Chern number 1. Each edge carries a single chiral mode. There is no counter-propagating mode to scatter into, in either spin, so no disorder can backscatter it as long as the bulk states at E_F are localised.
- The $G = 2$ phase is different: each edge carries two counter-propagating modes, one per spin, and only s_z conservation keeps them apart.
- Tilting the moments shows the difference. With $M_z = 0.8$ the in-plane components couple the spins. On the thesis map at $M_z = 0.8$ (same-g, $n_I = 0.2$, 56 configurations), the $G = 2$ plateau shrinks from 23 % to 3.8 % of the map, while the $G = 1$ plateau keeps 13.9 % (against 26 % at $M_z = 1$).

Opposite-g.

- The mean field is a spin-dependent mass, so the spin-up block does change sign, at $J_I \simeq 0.40$ for $n_I = 0.4$. The fluctuations delay this inversion relative to the mean field (0.34).

- In Fig. 5, $T_\uparrow = 1$ fills the corrected topological gap of the spin-up block: the wedge between its two band edges, which closes at $J_I \approx 0.41$, $E_F \approx 0.02$. $T_\uparrow = 0$ fills the trivial gap that opens beyond.
- The $G = 1$ phase is the region where the spin-up gap is trivial while spin down still conducts.
- The spin-down block becomes more topological, but at $n_I = 0.4$ its gap fills with impurity states for $J_I \gtrsim 0.45$. The exact broadening at $E_F = 0.02$ rises from 0.001 at $J_I = 0.35$ to 0.013, 0.024 and 0.048 at $J_I = 0.45$, 0.5 and 0.6. At $W = 64$, $T_\downarrow$ drops to 0.89, 0.60 and 0.32.
- In the paper's 64-site ribbons, the $G = 1$ window is therefore narrow at this concentration. It widens in wider ribbons, where both channels survive longer (Sec. 6).

8 What changes for the paper

Unchanged.

- The numerical results: conductance maps (Fig. 1), local currents (Fig. 2) and local densities of states (Fig. 3).
- The main conclusion: a disorder-induced $G = 1$ Chern-insulator phase for both couplings, carried by the spin-down edge state and caused by the suppression of one spin channel through selective scattering and localisation.

Changed.

- The sign of the mass equations of the SCBA, and the weight v_z.
- Fig. 4 and the paragraphs that interpret it:
  - same-g: the spin-down mass never changes sign, whereas the published lines invert it at $J_I \approx 1.0$–1.2. The plateau begins where the spin-up states at E_F stop conducting, after E_F has entered the spin-up band, shifted by the exchange field. The spin-up block becomes trivial only at stronger disorder, through a resonance beyond the Born approximation;
  - opposite-g: the spin-up inversion lies at $J_I \simeq 0.40$ for $n_I = 0.4$, and the left edge of the $G = 1$ region in the maps is set by the ribbon width.

Draft addendum.

> In the self-consistent Born analysis of Sec. IV B, the definition of the renormalised parameters should read $\mathrm{Re}\,\Sigma = \mathrm{diag}[(\bar{m}_\uparrow - m_\uparrow)\tau_z - (\bar{\mu}_\uparrow - \mu_\uparrow)\tau_0,\ (\bar{m}_\downarrow - m_\downarrow)\tau_z - (\bar{\mu}_\downarrow - \mu_\downarrow)\tau_0]$, so that the mass equations carry a plus sign in front of the sum. In addition, the longitudinal weight is $v_z = (1 - n_I)M_z^2 + (1 - M_z)^2/3$. The corrected Born phase diagrams are shown below.
>
> With the correct sign, the fluctuations renormalise both masses towards the topological side, as in the topological Anderson insulator, and the same-g spin-down mass does not change sign. The $G = 1$ phase of the same-g case arises because the spin-dependent shift of the bands brings the Fermi level into the spin-up band, whose states are localised by the disorder. At stronger disorder, the spin-up block becomes trivial through resonant scattering, beyond the Born approximation. Meanwhile the spin-down block keeps a topological gap and its chiral edge state, as shown in Fig. 2. In the opposite-g case, the spin-up inversion occurs at $J_I \approx 0.40$ for $n_I = 0.4$.
>
> The numerical results and the conclusions of the paper are unchanged.

A Methods and reproducibility

All calculations use the Rust toolbox `caroli`, which reproduces the thesis programs on their own test cases. Each item below is an example, run with

```
cargo run --release -p caroli-validation --example <name>
```

- `born_lines`: published and corrected Born parameters, on the grids of the published maps.
- `born_self_energy`: exact renormalised masses (Sec. 4.2).
- `born_spin_channels`: spin-resolved transmissions of Table 3.
- `born_wide_maps`: spin-resolved maps and cuts for $W = 64, 128, 256$.

The figures come from the plotting scripts `plot_born_phase.py`, `plot_born_masses.py` and `plot_born_wide.py` in the same folder.

Chebyshev expansion of the resolvent. With $\tilde{H} = (H - c)/h$ rescaled into $(-1, 1)$, $\tilde{z} = (E + \mathrm{i}\eta - c)/h$, $s = \sqrt{\tilde{z}^2 - 1}$ and $t = \tilde{z} - s$ chosen with $|t| < 1$,

$$\langle\phi|(E + \mathrm{i}\eta - H)^{-1}|\phi\rangle = \frac{1}{h\,s}\Big[\mu_0 + 2\sum_{n\geq 1} t^n \mu_n\Big], \qquad \mu_n = \langle\phi|T_n(\tilde{H})|\phi\rangle. \tag{14}$$

The moments are obtained two per product with $\tilde{H}$, from $\mu_{2n} = 2\langle T_n|T_n\rangle - \mu_0$ and $\mu_{2n+1} = 2\langle T_n|T_{n+1}\rangle - \mu_1$. The series is truncated where $|t|^n < 10^{-10}$.

Spin-resolved transport. With $M_z = 1$ each block is a two-orbital BHZ model with its own disorder. Its transmission is computed by recursive Green's functions, with the lead self-energies from the propagating and evanescent modes. The blocks reproduce the four-orbital calculation to 10^{-12}.

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